

BACKTEST PARAMETER COMPARISON

Screening yield, confirmation coverage, and parameter sensitivity

Evaluation window: 2022-03-01 to 2026-04-30 | Parameter sets: 192 | Compact outcomes: 324

Key conclusion: volume-ratio threshold is the dominant driver of screening yield. The broadest configuration selected 14 A-F and 1 A-H securities, while stricter volume settings reduced the candidate pool rapidly. This report compares screening coverage, not realized investment-return quality.

Parameter Choice Reference

- Six selectable parameters form the 192-combination grid. Daily and weekly moving-average tolerances remain fixed across every parameter set.

Parameter	Conditions	Choices	Mode	Meaning
Annual growth %	A	5%, 10%	Selectable	Minimum annual revenue and operating-income growth.
Quarterly growth %	B	5%, 10%	Selectable	Minimum quarterly revenue and operating-income growth.
Annual periods	A	2, 3 years	Selectable	Number of latest annual periods that must pass.
Quarterly periods	B	2, 3, 4 quarters	Selectable	Number of latest quarterly periods that must pass.
Volume ratio	C / D	2x, 3x, 4x, 5x	Selectable	Required volume-surge multiple.
Volume surge days	D	2, 3 days	Selectable	Minimum surge days in the recent-volume window.
Daily MA tolerance	E	1%	Fixed	Allowed daily MA20/MA50/MA100 alignment tolerance.
Weekly MA tolerance	G	2%	Fixed	Allowed weekly MA5/MA10/MA30 alignment tolerance.

Executive Summary

- The run completed successfully across all 192 parameter combinations with no duplicate outcomes, missing core prices, invalid dates, or high/low inconsistencies.
- A-F produced 304 parameter-security outcomes across 100 parameter sets and 14 unique securities.

- A-H produced 20 outcomes across 20 parameter sets and 1 unique securities; weekly confirmation narrows the candidate universe.
- Lower volume-ratio thresholds and shorter fundamental-history requirements produce the largest candidate pools.
- Return, median return, win rate, and drawdown ranking are not present in the execution log; those metrics should be queried from `backtest\_selection\_outcome` before selecting a final strategy.

Validated Run Overview

Screen	Outcomes	Sets with results	Unique stocks	First signal	Last entry
A-F	304	100	14	2022-03-01	2026-04-30
A-H	20	20	1	2025-08-05	2025-08-15

Average selected stocks per parameter set

A-F and A-H screening yield by selected parameter value

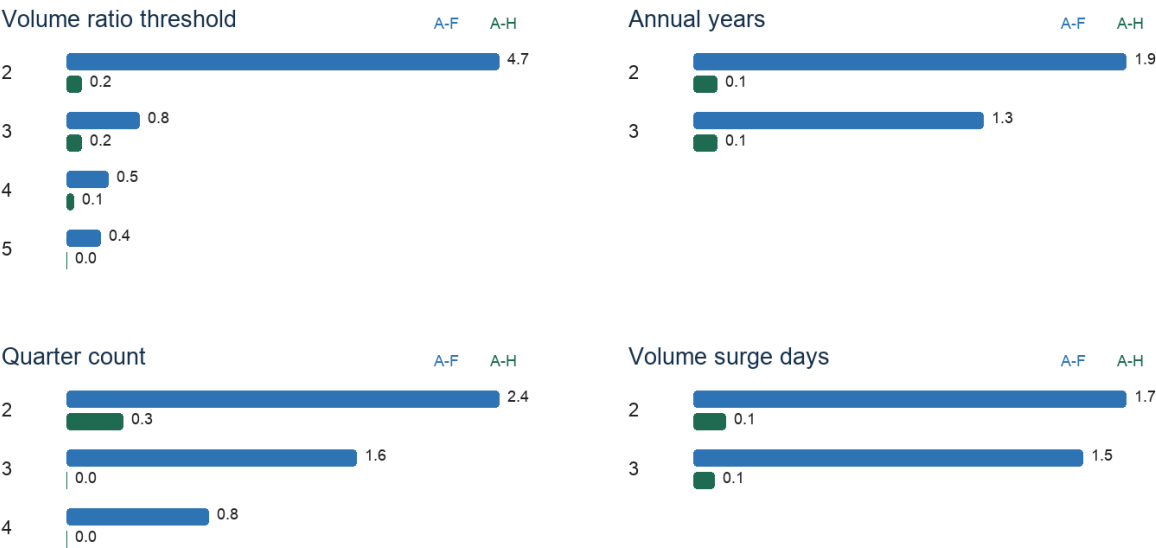


Figure 1. Average screening yield by parameter choice.

Parameter Sensitivity

Factor	Compared values	Avg A-F yield	Avg A-H yield
Annual growth threshold	5 -> 10	2.3 -> 0.9	0.2 -> 0.0
Quarterly growth threshold	5 -> 10	2.0 -> 1.2	0.1 -> 0.1

Annual periods	2 -> 3	1.9 -> 1.3	0.1 -> 0.1
Quarterly periods	2 -> 4	2.4 -> 0.8	0.3 -> 0.0
Volume ratio threshold	2 -> 5	4.7 -> 0.4	0.2 -> 0.0
Volume surge days	2 -> 3	1.7 -> 1.5	0.1 -> 0.1

Interpretation

- Volume ratio: moving from 2 to 5 changes average A-F yield from 4.7 to 0.4 and average A-H yield from 0.2 to 0.0.
- Annual-period requirement: moving from 2 to 3 changes average A-F yield from 1.9 to 1.3 and average A-H yield from 0.1 to 0.1.
- Quarterly-period requirement: moving from 2 to 4 changes average A-F yield from 2.4 to 0.8 and average A-H yield from 0.3 to 0.0.
- Volume-surge-day requirement: moving from 2 to 3 changes average A-F yield from 1.7 to 1.5 and average A-H yield from 0.1 to 0.1.

Highest-Coverage Parameter Sets

ID	Growth A/Q	Periods A/Q	Volume	A-F	A-H	Total
1	5/5	2/2	2x/2d	14	1	15
2	5/5	2/2	2x/3d	13	1	14
9	5/5	2/3	2x/2d	10	0	10
25	5/5	3/2	2x/2d	8	1	9
26	5/5	3/2	2x/3d	8	1	9
49	5/10	2/2	2x/2d	8	1	9
10	5/5	2/3	2x/3d	9	0	9
50	5/10	2/2	2x/3d	7	1	8
33	5/5	3/3	2x/2d	7	0	7
34	5/5	3/3	2x/3d	7	0	7

Lowest-Coverage Parameter Sets

- These configurations produced the fewest combined A-F and A-H selections. Their sparse samples are useful for understanding selectivity, but are too small for reliable return-performance conclusions.

ID	Growth A/Q	Periods A/Q	Volume	A-F	A-H	Total
19	5/5	2/4	3x/2d	0	0	0
20	5/5	2/4	3x/3d	0	0	0
21	5/5	2/4	4x/2d	0	0	0
22	5/5	2/4	4x/3d	0	0	0
23	5/5	2/4	5x/2d	0	0	0
24	5/5	2/4	5x/3d	0	0	0
43	5/5	3/4	3x/2d	0	0	0
44	5/5	3/4	3x/3d	0	0	0
45	5/5	3/4	4x/2d	0	0	0
46	5/5	3/4	4x/3d	0	0	0

Recommended Comparison Set

Role	ID	Configuration	A-F / A-H	Use
Broad discovery	1	5%/5%, 2 annual, 2 quarters, 2x volume, 2 surge days	14 / 1	Largest sample; suitable baseline.
Moderate volume	3	5%/5%, 2 annual, 2 quarters, 3x volume, 2 surge days	5 / 1	Stricter volume comparator.
Strict volume	7	5%/5%, 2 annual, 2 quarters, 5x volume, 2 surge days	2 / 0	High-conviction, smaller-sample comparator.
Durability focus	41	5%/5%, 3 annual, 4 quarters, 2x volume, 2 surge days	5 / 0	Longer fundamental-history comparator.

Decision Guidance

- Do not label the highest-coverage parameter set as the best-performing investment strategy. Coverage and return quality answer different questions.

- Require a minimum sample size before ranking return performance. A practical initial threshold is at least 20 A-F outcomes or at least 10 A-H outcomes.
- Compare mean and median return together. A large gap often indicates that a few extreme winners dominate the average.
- Use win rate and maximum drawdown alongside returns; parameter sets with high average returns but severe drawdowns may be unsuitable.
- Evaluate A-F and A-H separately. A-H is a stricter weekly-confirmed strategy and should not be pooled with A-F.

Required Return-Performance Query

To complete an investment-performance ranking, export the following aggregate query from Supabase and use it for a second report:

```
SELECT p.parameter_set_id, p.parameter_set_name, o.screen_type, COUNT(*) AS sample_size,
ROUND(AVG(o.return_pct), 2) AS avg_return_pct,
ROUND(PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY o.return_pct)::numeric, 2) AS median_return_pct,
ROUND(100.0 * COUNT(*) FILTER (WHERE o.return_pct > 0) / COUNT(*), 2) AS win_rate_pct,
ROUND(AVG(o.max_drawdown_pct), 2) AS avg_drawdown_pct,
ROUND(AVG(o.max_return_pct), 2) AS avg_max_return_pct
FROM backtest_parameter_set p JOIN backtest_selection_outcome o USING (parameter_set_id)
GROUP BY p.parameter_set_id, p.parameter_set_name, o.screen_type
ORDER BY o.screen_type, sample_size DESC;
```

Scope and Limitations

- This report uses the validated execution log and compares selection yield, not realized portfolio returns.
- The same security may appear in many parameter sets, so outcome rows are not independent observations.
- Fundamental histories use `datadate <= signal\_date`; actual public filing-availability dates are unavailable, so some look-ahead-bias risk remains.
- No transaction costs, liquidity constraints, holding rules, or overlapping-position controls are included.